Vectors

3 September 2025

An Introduction to Vectors

Two points P and Q define a **vector** $\vec{v}$. The vector connects P and Q, and we write $\vec{v} = \vec{PQ}$. On the two-dimensional *plane* $\mathbb{R}^2$, we can represent a point P as a vector $\vec{OP}$ that connects the origin with the point P,

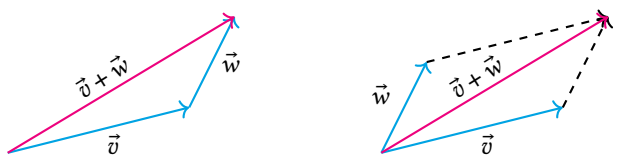
$$\vec{v} = \begin{pmatrix} x \\ y \end{pmatrix},$$

where x and y are the *components* of $\vec{v}$ given by the Cartesian coordinates of the point P. Similarly, in three-dimensional *space* $\mathbb{R}^3$, we can represent a point P as a vector $\vec{OP}$ that connects the origin with the point P,

$$\vec{v} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}.$$

Remark. For scientists, vectors are often used to indicate quantities that have both a *magnitude* and a *direction*, such as an object's displacement or a force applied. Note, however, that not all vectors have a magnitude and a direction: the *zero vector* $\vec{0}$ has length 0, and it has no specific direction.

We now introduce some operations on vectors. If $\vec{v}, \vec{w}$ are vectors positioned such that the tail of $\vec{w}$ is at the head of $\vec{v}$, then the **sum** $\vec{v} + \vec{w}$ is the vector from the tail of $\vec{v}$ to the head of $\vec{w}$. Alternatively, if we place $\vec{v}$ and $\vec{w}$ such that their tails are at the same point, then $\vec{v} + \vec{w}$ is the diagonal of the parallelogram with sides $\vec{v}$ and $\vec{w}$.



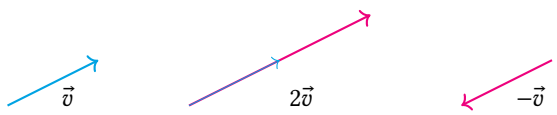
In Cartesian coordinates, the sum is computed component-wise: in $\mathbb{R}^2$, for instance, we have

$$\vec{v} + \vec{w} = \begin{pmatrix} x_1 \\ y_1 \end{pmatrix} + \begin{pmatrix} x_2 \\ y_2 \end{pmatrix} = \begin{pmatrix} x_1 + x_2 \\ y_1 + y_2 \end{pmatrix}.$$

Similarly, in xyz-space, we have

$$\vec{v} + \vec{w} = \begin{pmatrix} x_1 \\ y_1 \\ z_1 \end{pmatrix} + \begin{pmatrix} x_2 \\ y_2 \\ z_2 \end{pmatrix} = \begin{pmatrix} x_1 + x_2 \\ y_1 + y_2 \\ z_1 + z_2 \end{pmatrix}.$$

Given a scalar λ, the **scalar multiple** $\lambda\vec{v}$ is the vector whose length is $|\lambda|$ times the length of $\vec{v}$, and whose direction is the same as $\vec{v}$ if $\lambda > 0$ and opposite to $\vec{v}$ if $\lambda < 0$.



We may use Cartesian coordinates to compute the scalar multiple of a vector $\vec{v}$: for instance, in $\mathbb{R}^2$,

$$\lambda\vec{v} = \lambda \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \lambda x \\ \lambda y \end{pmatrix}.$$

If either $\lambda = 0$ or $\vec{v} = \vec{0}$, then $\lambda\vec{v} = \vec{0}$. With these operations, we can define the **difference** $\vec{v} - \vec{w}$ of two vectors: this is best seen as

$$\vec{v} - \vec{w} = \vec{v} + (-1)\,\vec{w}.$$

Note that, given two points with position vectors $\vec{OP}$ and $\vec{OQ}$, the vector $\vec{PQ}$ is thus given by

$$\vec{PQ} = \vec{OQ} - \vec{OP}.$$

Geometry and Distance

The **magnitude** (or **length**) of a vector $\vec{v} = \vec{PQ}$, denoted by $|\vec{v}|$, is the distance from P to Q. For a vector $\vec{v}$ in $\mathbb{R}^2$ with Cartesian components x, y, the Pythagorean Theorem tells us that

$$|\vec{v}| = \left| \begin{pmatrix} x \\ y \end{pmatrix} \right| = \sqrt{x^2 + y^2}.$$

In $\mathbb{R}^3$, we have

$$|\vec{v}| = \left| \begin{pmatrix} x \\ y \\ z \end{pmatrix} \right| = \sqrt{x^2 + y^2 + z^2}.$$

A vector of length 1 is called a **unit vector**. For $\vec{v} \neq \vec{0}$, the vector $\vec{v}/|\vec{v}|$ is a unit vector called the **direction vector** of $\vec{v}$.

Example. Consider the vector

$$\vec{v} = \begin{pmatrix} 3 \\ 4 \\ 12 \end{pmatrix}.$$

Its magnitude is given by

$$|\vec{v}| = \sqrt{3^2 + 4^2 + 12^2} = 13.$$

We can thus determine a unit vector in the direction of $\vec{v}$:

$$\vec{u} = \frac{1}{|v|}\vec{v} = \begin{pmatrix} 3/13 \\ 4/13 \\ 12/13 \end{pmatrix}.$$

The **standard basis vectors** are a special set of unit vectors: in $\mathbb{R}^2$, they are given by

$$\vec{i} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \vec{j} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

These vectors are unit vectors, and any vector $\vec{v}$ in $\mathbb{R}^2$ can be written as a combination of standard basis vectors

$$\vec{v} = \begin{pmatrix} x \\ y \end{pmatrix} = x\,\vec{i} + y\,\vec{j}.$$

Similarly, the standard basis vectors in $\mathbb{R}^3$ are given by

$$\vec{i} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad \vec{j} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \quad \vec{k} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}.$$

Any vector $\vec{v}$ in space can be written as a combination of standard basis vectors

$$\vec{v} = \begin{pmatrix} x \\ y \\ z \end{pmatrix} = x\,\vec{i} + y\,\vec{j} + z\,\vec{k}.$$

We now define a special type of product between vectors that is deeply related to the notion of distance. The **dot product** of two vectors $\vec{v}, \vec{w}$ in $\mathbb{R}^2$ is a *scalar* given by

$$\vec{v} \cdot \vec{w} = \begin{pmatrix} x_1 \\ y_1 \end{pmatrix} \cdot \begin{pmatrix} x_2 \\ y_2 \end{pmatrix} = x_1x_2 + y_1y_2.$$

The dot product of two vectors $\vec{v}, \vec{w}$ in $\mathbb{R}^3$ is defined similarly:

$$\vec{v} \cdot \vec{w} = \begin{pmatrix} x_1 \\ y_1 \\ z_1 \end{pmatrix} \cdot \begin{pmatrix} x_2 \\ y_2 \\ z_2 \end{pmatrix} = x_1x_2 + y_1y_2 + z_1z_2.$$

A key observation to make is that the dot product determines the notion of distance: if we compute the dot product a vector $\vec{v}$ with itself, we have

$$\vec{v} \cdot \vec{v} = \begin{pmatrix} x \\ y \\ z \end{pmatrix} \cdot \begin{pmatrix} x \\ y \\ z \end{pmatrix} = x^2 + y^2 + z^2.$$

In particular, we have

$$|\vec{v}|^2 = \vec{v} \cdot \vec{v}.$$

In addition, we have the following properties of the dot product that can easily be shown with the dot product formula.

Proposition 1. *Let $\vec{u}, \vec{v}, \vec{w}$ be vectors, and let λ be a scalar. Then,*

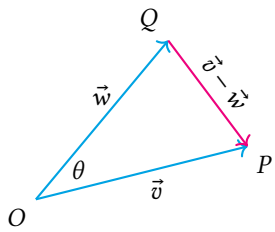
1. $\vec{v} \cdot \vec{w} = \vec{w} \cdot \vec{v}$
2. $\vec{u} \cdot (\vec{v} + \vec{w}) = \vec{u} \cdot \vec{v} + \vec{u} \cdot \vec{w}$
3. $\lambda\,(\vec{v} \cdot \vec{w}) = (\lambda\vec{v}) \cdot \vec{w} = \vec{v} \cdot (\lambda\vec{w})$
4. $\vec{0} \cdot \vec{v} = 0$

With the dot product, we can begin to examine other geometric objects such as the **angle** between two vectors.

Theorem 2. *Let θ be the angle between two vectors $\vec{v}$ and $\vec{w}$. Then,*

$$\vec{v} \cdot \vec{w} = |\vec{v}|\,|\vec{w}| \cos\theta.$$

Proof. It is clear that the equality holds when either $\vec{v}$ or $\vec{w}$ is the zero vector. We thus suppose that $\vec{v}, \vec{w} \neq \vec{0}$, and we use geometry: let $\vec{v} = \vec{OP}$ and $\vec{w} = \vec{OQ}$, and consider the triangle OPQ.



By the Law of Cosines on the triangle OPQ, we get

$$|PQ|^2 = |OP|^2 + |OQ|^2 - 2\,|OP|\,|OQ| \cos\theta.$$

In particular, we have

$$|\vec{v} - \vec{w}|^2 = |\vec{v}|^2 + |\vec{w}|^2 - 2\,|\vec{v}|\,|\vec{w}| \cos\theta.$$

We can expand the term on the left-hand side with the dot product:

$$\begin{aligned}
|\vec{v} - \vec{w}|^2 &= (\vec{v} - \vec{w}) \cdot (\vec{v} - \vec{w}) \\
&= (\vec{v} - \vec{w}) \cdot \vec{v} - (\vec{v} - \vec{w}) \cdot \vec{w} \\
&= \vec{v} \cdot \vec{v} - \vec{w} \cdot \vec{v} - \vec{v} \cdot \vec{w} + \vec{w} \cdot \vec{w} \\
&= |\vec{v}|^2 - 2\vec{v} \cdot \vec{w} + |\vec{w}|^2 .
\end{aligned}$$

Hence,

$$|\vec{v}|^2 - 2\vec{v} \cdot \vec{w} + |\vec{w}|^2 = |\vec{v}|^2 + |\vec{w}|^2 - 2\,|\vec{v}|\,|\vec{w}| \cos\theta.$$

We thus arrive at the desired result:

$$\vec{v} \cdot \vec{w} = |\vec{v}|\,|\vec{w}| \cos\theta.$$

□

This motivates the definition of the **angle** between two non-zero vectors: it is the unique value $\theta \in [0, \pi]$ that satisfies

$$\cos\theta = \frac{\vec{v} \cdot \vec{w}}{|\vec{v}|\,|\vec{w}|}.$$

Example. Let

$$\vec{v} = \begin{pmatrix} 2 \\ 2 \\ -1 \end{pmatrix}, \quad \vec{w} = \begin{pmatrix} 5 \\ -3 \\ 2 \end{pmatrix}.$$

We can determine the angle between them: we compute

$$\begin{aligned}
|\vec{v}| &= \sqrt{2^2 + 2^2 + (-1)^2} = 3, \\
|\vec{w}| &= \sqrt{5^2 + (-3)^2 + 2^2} = \sqrt{38}, \\
\vec{v} \cdot \vec{w} &= 2 \cdot 5 + 2 \cdot -3 + -1 \cdot 2 = 2.
\end{aligned}$$

Hence,

$$\cos\theta = \frac{\vec{v} \cdot \vec{w}}{|\vec{v}|\,|\vec{w}|} = \frac{2}{3\sqrt{38}} \quad \Longrightarrow \quad \theta = \cos^{-1}\left(\frac{2}{3\sqrt{38}}\right) \approx 1.46.$$

Note that two nonzero vectors $\vec{v}, \vec{w}$ are **perpendicular**, or **orthogonal**, to one another if the angle between them is $\theta = \pi/2$: then,

$$\vec{v} \cdot \vec{w} = |\vec{v}|\,|\vec{w}| \cos(\pi/2) = 0.$$

We thus conclude that $\vec{v}, \vec{w}$ *are orthogonal if and only if* $\vec{v} \cdot \vec{w} = 0$.

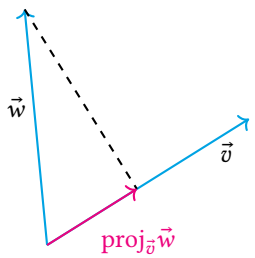
Projections

The relationship between the angle and the dot product provides us with another geometric notion: if $\vec{u}$ is a unit vector, then $\vec{u} \cdot \vec{v}$ is the the length of the projection of $\vec{v}$ onto $\vec{u}$—that is, $\vec{u} \cdot \vec{v}$ is the length of the 'shadow' that $\vec{v}$ casts in the direction of $\vec{u}$ when both vectors' tails start at the same point. Now, given two arbitrary vectors $\vec{v}$ and $\vec{w}$, we have

$$|\vec{w}| \cos\theta = \frac{\vec{v} \cdot \vec{w}}{|\vec{v}|} = \frac{\vec{v}}{|\vec{v}|} \cdot \vec{w},$$

which gives us the component of $\vec{v}$ along $\vec{u}$. From this, we can construct the **projection vector**

$$\text{proj}_{\vec{v}}\vec{w} = \left(\frac{\vec{v} \cdot \vec{w}}{|\vec{v}|}\right) \frac{\vec{v}}{|\vec{v}|} = \left(\frac{\vec{v} \cdot \vec{w}}{|\vec{v}|^2}\right) \vec{v}.$$



Example. Let us find the projection of $\vec{w} = \vec{i} + \vec{j} + 2\vec{k}$ onto $\vec{v} = -2\vec{i} + 3\vec{j} + \vec{k}$. We can simply plug these in directly to the formula: observe that

$$\begin{aligned} \vec{v} \cdot \vec{w} &= (-2 \cdot 1) + (3 \cdot 1) + (1 \cdot 2) = 3, \\ |\vec{v}|^2 = \vec{v} \cdot \vec{v} &= (-2)^2 + 3^2 + 1^2 = 14. \end{aligned}$$

Hence,

$$\text{proj}_{\vec{v}}\vec{w} = \frac{3}{14}\vec{v} = \frac{3}{14}\begin{pmatrix} -2 \\ 3 \\ 1 \end{pmatrix} = \begin{pmatrix} -3/7 \\ 9/14 \\ 3/14 \end{pmatrix}.$$

The Cross Product

Given two non-zero vectors $\vec{v}$ and $\vec{w}$, it is often useful to be able to find a third non-zero vector $\vec{u}$ that is perpendicular to both $\vec{v}$ and $\vec{w}$. This is given to us by the **cross product**, which is uniquely defined in $\mathbb{R}^3$ as the *vector*

$$\vec{v} \times \vec{w} = \begin{pmatrix} x_1 \\ y_1 \\ z_1 \end{pmatrix} \times \begin{pmatrix} x_2 \\ y_2 \\ z_2 \end{pmatrix} = \begin{pmatrix} y_1 z_2 - z_1 y_2 \\ z_1 x_2 - x_1 z_2 \\ x_1 y_2 - y_1 x_2 \end{pmatrix}.$$

Theorem 3. *The vector $\vec{v} \times \vec{w}$ is perpendicular to both $\vec{v}$ and $\vec{w}$.*

Example. The cross product of the vectors $\vec{v} = \vec{i} + 3\vec{j} + 4\vec{k}$ and $\vec{w} = 2\vec{i} + 7\vec{j} - 5\vec{k}$ is given by

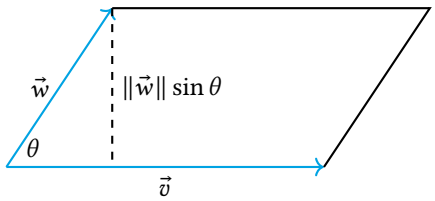
$$\begin{pmatrix} 1 \\ 3 \\ 4 \end{pmatrix} \times \begin{pmatrix} 2 \\ 7 \\ -5 \end{pmatrix} = \begin{pmatrix} -15 - 28 \\ 8 - (-5) \\ 7 - 6 \end{pmatrix} = \begin{pmatrix} -43 \\ 13 \\ 1 \end{pmatrix}.$$

Let us now gain some geometric intuition for the magnitude of the cross product.

Theorem 4. *If $\theta \in [0, \pi]$ is the angle between two vectors $\vec{v}$ and $\vec{w}$, then the magnitude of the cross product is given by*

$$|\vec{v} \times \vec{w}| = |\vec{v}|\,|\vec{w}| \sin\theta.$$

This theorem allows for a geometric interpretation of the magnitude of the cross product: $|\vec{v} \times \vec{w}|$ is equal to the area of the parallelogram determined by $\vec{v}$ and $\vec{w}$.



We end with some properties of the cross product.

Proposition 5. *Let $\vec{u}, \vec{v}, \vec{w}$ be vectors in $\mathbb{R}^3$, and let λ be a scalar. Then,*

1. $\vec{v} \times \vec{w} = -\vec{w} \times \vec{v}$
2. $(c\vec{v}) \times \vec{w} = c\,(\vec{v} \times \vec{w}) = \vec{v} \times c\vec{w}$
3. $\vec{u} \times (\vec{v} + \vec{w}) = \vec{u} \times \vec{v} + \vec{u} \times \vec{w}$
4. $(\vec{v} + \vec{w}) \times \vec{u} = \vec{v} \times \vec{u} + \vec{w} \times \vec{u}$
5. $\vec{u} \cdot (\vec{v} \times \vec{w}) = (\vec{u} \times \vec{v}) \cdot \vec{w}$
6. $\vec{u} \times (\vec{v} \times \vec{w}) = (\vec{u} \cdot \vec{w})\,\vec{v} - (\vec{u} \cdot \vec{v})\,\vec{w}$

All of these results can be proven by direct expansion using the definition of the cross product.

Example. Consider the vectors

$$\vec{u} = \begin{pmatrix} 1 \\ 4 \\ -7 \end{pmatrix}, \quad \vec{v} = \begin{pmatrix} 2 \\ -1 \\ 4 \end{pmatrix}, \quad \vec{w} = \begin{pmatrix} 0 \\ -9 \\ 18 \end{pmatrix}.$$

Let us show that the three vectors must all lie on the same plane. A vector perpendicular to both $\vec{u}$ and $\vec{v}$ (and thus perpendicular to the plane containing them) is given by

$$\vec{u} \times \vec{v} = \begin{pmatrix} 1 \\ 4 \\ -7 \end{pmatrix} \times \begin{pmatrix} 2 \\ -1 \\ 4 \end{pmatrix} = \begin{pmatrix} 16 - 7 \\ -14 - 4 \\ -1 - 8 \end{pmatrix} = \begin{pmatrix} 9 \\ -18 \\ -9 \end{pmatrix}.$$

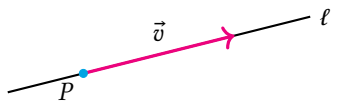
Observe that, if $\vec{w}$ were to lie on the same plane as $\vec{u}$ and $\vec{v}$, then it must be perpendicular to the above vector as well: we can check that

$$\vec{w} \cdot (\vec{u} \times \vec{v}) = \begin{pmatrix} 0 \\ -9 \\ 18 \end{pmatrix} \cdot \begin{pmatrix} 9 \\ -18 \\ -9 \end{pmatrix} = 0 \cdot 9 + (-9) \cdot (-18) + 18 \cdot (-9) = 0.$$

Hence, all three vectors must lie on the same plane.

Equations of Lines and Planes

Equipped with the language of vectors, we can begin to describe some objects in space. A **line** ℓ is determined when we know a point P on the line, as well as its direction.



Given a point $P\,(x_0, y_0, z_0)$ on a line ℓ, and a vector $\vec{v} = a\,\vec{i} + b\,\vec{j} + c\,\vec{k}$ parallel to ℓ, a **vector equation** of ℓ is given by

$$\vec{r}\,(t) = \vec{OP} + t\vec{v} = \begin{pmatrix} x_0 \\ y_0 \\ z_0 \end{pmatrix} + t \begin{pmatrix} a \\ b \\ c \end{pmatrix} \text{ for } t \in \mathbb{R}.$$

Each value of the *parameter* t gives a position vector $\vec{r}$ that determines a point on ℓ. Since the vector equation determines the (x, y, z) coordinates of each point in the line, we also have the **parametric equations**

$$x = x_0 + at, \quad y = y_0 + bt, \quad z = z_0 + ct, \quad t \in \mathbb{R}.$$

From the parametric equations, we can determine equivalent expressions for t that give us yet another way of representing the same line:

$$\frac{x - x_0}{a} = \frac{y - y_0}{b} = \frac{z - z_0}{c}.$$

These are called the **symmetric equations** of ℓ.

Example. Consider the line ℓ that contains the point $P\,(2, 4, -3)$ and $Q\,(3, -1, 1)$. Let us determine some representations of this line: observe that a vector parallel to the line is given by

$$\vec{PQ} = \begin{pmatrix} 3 \\ -1 \\ 1 \end{pmatrix} - \begin{pmatrix} 2 \\ 4 \\ -3 \end{pmatrix} = \begin{pmatrix} 1 \\ -5 \\ 4 \end{pmatrix}.$$

Thus, a vector equation of the line is given by

$$\vec{r} = \vec{OP} + t\vec{PQ} = \begin{pmatrix} 2 \\ 4 \\ -3 \end{pmatrix} + t \begin{pmatrix} 1 \\ -5 \\ 4 \end{pmatrix}, \quad t \in \mathbb{R}.$$

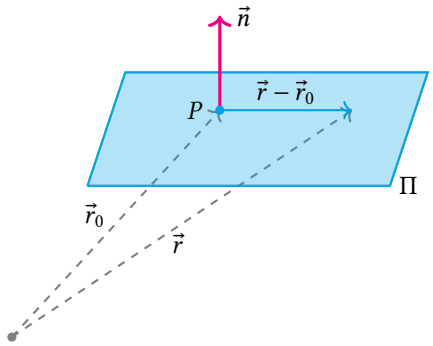
The parametric equations are hence

$$x = 2 + t, \quad y = 4 - 5t, \quad z = -3 + 4t,$$

and from this we have the symmetric equations

$$x - 2 = \frac{y - 4}{-5} = \frac{z + 3}{4}.$$

A **plane** in space is determined by a point on the plane and a vector orthogonal to the plane. This perpendicular vector is called a **normal vector**. We now determine an equation for the plane: let $\vec{r}_0$ be the position vector of a point $P\,(x_0, y_0, z_0)$ that lies on a plane Π, and let $\vec{n} = a\,\vec{i} + b\,\vec{j} + c\,\vec{k}$ be a normal vector to Π.



Given an arbitrary point (x, y, z) with position vector $\vec{r}$ on the plane, the vector $\vec{r} - \vec{r}_0$ must be orthogonal to $\vec{n}$. We can thus use the dot product to obtain a **vector equation** of the plane:

$$\vec{n} \cdot (\vec{r} - \vec{r}_0) = 0 \quad \Longleftrightarrow \quad \begin{pmatrix} a \\ b \\ c \end{pmatrix} \cdot \left[\begin{pmatrix} x \\ y \\ z \end{pmatrix} - \begin{pmatrix} x_0 \\ y_0 \\ z_0 \end{pmatrix} \right] = 0.$$

Expanding this equation out yields the **Cartesian equation** of Π,

$$a\,(x - x_0) + b\,(y - y_0) + c\,(z - z_0) = 0.$$

Example. Let us find the equation of the plane that contains the points $P\,(1, 3, 2)$, $Q\,(3, -1, 6)$, and $R\,(5, 2, 0)$. To do this, we need a point on the plane (of which we have three) and a normal vector. Observe that the plane must contain the vectors

$$\vec{PQ} = \begin{pmatrix} 3-1 \\ -1-3 \\ 6-2 \end{pmatrix} = \begin{pmatrix} 2 \\ -4 \\ 4 \end{pmatrix}, \quad \vec{PR} = \begin{pmatrix} 5-1 \\ 2-3 \\ 0-2 \end{pmatrix} = \begin{pmatrix} 4 \\ -1 \\ -2 \end{pmatrix}.$$

Their cross product must be orthogonal to the plane, and may thus be taken as its normal vector:

$$\vec{PQ} \times \vec{PR} = \begin{pmatrix} 2 \\ -4 \\ 4 \end{pmatrix} \times \begin{pmatrix} 4 \\ -1 \\ -2 \end{pmatrix} = \begin{pmatrix} 8-(-4) \\ 16-(-4) \\ -2-(-16) \end{pmatrix} = \begin{pmatrix} 12 \\ 20 \\ 14 \end{pmatrix}.$$

With the point $P\,(1, 3, 2)$, a vector equation of the plane is given by

$$\begin{pmatrix} 12 \\ 20 \\ 14 \end{pmatrix} \cdot \left[\begin{pmatrix} x \\ y \\ z \end{pmatrix} - \begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix} \right] = 0.$$

This yields the Cartesian equation

$$12\,(x-1) + 20\,(y-3) + 14\,(z-2) = 0.$$

Collecting terms, we may simplify this to

$$12x + 20y + 14z = 100 \quad \Longrightarrow \quad 6x + 10y + 7z = 50.$$

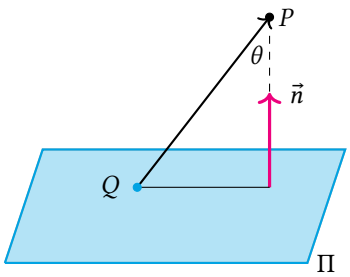
Distances

We end with some distance formulae in that allow us to measure lengths in space.

1. Let P be a point, and let Π be the plane with normal vector $\vec{n}$ passing through Q. Then, the distance between P and Π is given by

$$d(P, \Pi) = \frac{\left|\vec{n} \cdot \vec{PQ}\right|}{|\vec{n}|}.$$

   To see this, we draw a picture:

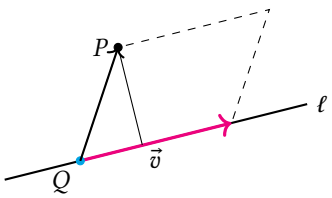


   The distance is simply the length of the projection of $\vec{PQ}$ onto $\vec{n}$.

2. Let P be a point, and let ℓ be the line through Q in the direction of $\vec{v}$. Then,

$$d(P, \ell) = \frac{\left|\vec{v} \times \vec{PQ}\right|}{|\vec{v}|}.$$

   To see this, we use geometry.



   Recall that the cross product gives the area of a parallelogram: in particular, the height of a parallelogram is simply its area divided by the length of its base.

Exercise. Show that the distance between two parallel planes

$$\Pi : \vec{n} \cdot \vec{r} = d, \qquad \Sigma : \vec{n} \cdot \vec{r} = e$$

is given by

$$d(\Pi, \Sigma) = \frac{|d - e|}{|\vec{n}|}.$$

Hence, show that the distance between skew lines

$$\ell : \vec{r} = P + t\vec{v}, \qquad m : \vec{r} = Q + t\vec{w}$$

is given by

$$d(\ell, m) = \frac{\left|\vec{PQ} \cdot (\vec{v} \times \vec{w})\right|}{|\vec{v} \times \vec{w}|}.$$